

NEWPORT HARBOR, RI, USA

WMO: 997285

Lat: 41.504N

Lon: 71.326W

Elev: 33

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.5	16.7	N/A	N/A	N/A	N/A	N/A	N/A	31.2	37.0	27.8	36.2	14.4	0	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	81.1	N/A	78.5	N/A	76.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	7.3	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.8	23.5	20.2	DB	9.1	88.6	4.8	3.7	5.7	91.3	2.8	93.5	0.1	95.6	-3.4	98.3	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	31.8	32.6	38.3	47.9	56.1	65.2	72.2	71.5	65.8	57.1	47.0	38.2
	DBStd	15.37	8.92	7.27	7.07	5.13	5.16	4.88	3.42	3.29	4.46	6.00	7.18	7.61
	HDD50	2037	563	486	365	98	6	0	0	0	0	13	138	368
	HDD65	5279	1028	906	828	513	280	55	1	1	45	253	539	830
	CDD50	2800	0	0	2	35	196	456	689	665	473	231	49	4
	CDD65	567	0	0	0	0	5	61	225	201	68	7	0	0
	CDH74	1239	0	0	0	0	18	108	589	441	74	9	0	0
	CDH80	127	0	0	0	0	3	12	71	38	2	1	0	0
Wind	WSAvg	9.8	10.9	10.9	11.3	10.7	9.6	8.8	8.2	7.7	8.2	10.1	10.7	10.9
Precipitation	PrecAvg	46.3	3.7	3.5	4.7	4.1	3.5	3.6	3.3	3.8	4.0	3.9	3.9	4.4
	PrecMax	56.1	6.2	6.5	15.2	8.0	7.4	10.7	10.1	7.8	9.3	13.0	6.7	7.2
	PrecMin	37.1	1.7	1.8	0.3	1.5	1.1	0.0	1.0	0.8	0.8	0.4	0.7	1.3
	PrecStd	5.3	1.2	1.3	3.0	1.7	1.3	2.6	1.8	1.8	1.9	2.6	1.6	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	50.2	57.8	68.2	76.0	81.2	86.1	83.8	79.3	73.9	63.6	57.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	51.1	46.8	53.0	60.5	69.5	76.9	81.7	80.4	76.3	69.8	61.1	55.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	48.0	44.4	49.9	57.3	66.1	74.0	79.1	77.9	73.8	67.4	59.4	52.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	44.9	42.4	47.4	54.9	63.3	71.7	77.2	76.4	72.1	65.6	57.9	50.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	11.2	10.8	10.8	10.7	10.1	9.5	8.8	8.9	9.4	9.5	10.0	10.5
		MCDBR	13.6	11.9	13.8	14.7	14.5	13.1	12.1	10.7	10.8	10.0	10.4	13.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.307	0.315	0.338	0.375	0.418	0.462	0.487	0.444	0.377	0.346	0.317	0.305	
	taud	2.504	2.481	2.455	2.385	2.299	2.214	2.152	2.265	2.428	2.519	2.558	2.557	
	Ebn at Noon	270	285	289	284	273	260	252	260	272	270	265	262	
	Edh at Noon	23	28	32	36	41	45	47	41	32	26	22	21	
All-Sky Solar Radiation	RadAvg	594	873	1194	1541	1708	1867	1915	1721	1402	950	649	499	
	RadStd	46	86	87	165	163	185	100	90	103	86	46	49	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+173	+85

Nomenclature: See separate page